

Ansh Singhal

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SUMMARY

AI/ML Engineer with hands-on experience in building *RAG systems, LLM applications, hybrid retrieval pipelines, and fine-tuned models* for real-world tasks. Proven ability to deliver *scalable end-to-end ML solutions* and collaborate in *production-oriented deployment workflows*.

SKILLS

- **Programming Languages:** Python, Java, C++, SQL
- **Machine Learning & AI:** PyTorch, TensorFlow, Hugging Face Transformers, LangChain, LlamaIndex
- **Backend & Web Frameworks:** Spring Boot, Flask, FastAPI, Django, REST APIs, gRPC, Bootstrap
- **DevOps & MLOps Tools:** Docker, Kubernetes, Redis, Kafka, Celery, Git, Prometheus, Grafana, CI/CD
- **Cloud Platforms:** Google Cloud Platform (GCP), Microsoft Azure, Amazon Web Services (AWS)
- **Systems Design:** API Design, Observability, Scalable Microservices, Distributed Systems

EDUCATION

- **Bennett University** Uttar Pradesh, India
B.Tech in Computer Science (AI Specialization); CGPA: 9.65 2023 – 2027

EXPERIENCE

- **IntelOwl – Threat Intelligence Platform** Remote
Open Source Contributor Jan 2025 – May 2025
 - **Threat Analyzer Development:** Built multiple file- and event-based analyzers to extract actionable threat intelligence from malware samples and URLs.
 - **Containerization & APIs:** Containerized analyzers using **Docker** and exposed them via **FastAPI**, ensuring scalable, isolated execution.
 - **Django Backend:** Developed and maintained a **Django** backend exposing RESTful endpoints for OSINT enrichment, enabling easy integration.

PROJECTS

- **Agentic RAG – Legal Research Engine:** Built an **agentic RAG pipeline** for domain-specific legal retrieval with **97% accuracy** and **P95 latency ; 500ms**. Designed multi-agent routing via LangGraph and hybrid dense+sparse retrieval using Pinecone.
Highlights: 2-stage reranking (BGE + MiniLM), LegalSeg-based NLP, containerized FastAPI microservices on **AWS ECS**.
Tech: Python, FastAPI, LangGraph, Pinecone, Docker, AWS ECS
- **Flood-GAN – Physics-Aware Flood Mapping:** Designed a **physics-informed cGAN** for **SAR→Optical translation** to detect floods. Achieved **PSNR 31.25, SSIM 0.94, and L1 0.05**. Model outputs RGB + water segmentation maps with hydrology-consistent supervision.
Highlights: NDWI-consistent RGB generation, cloud-masked losses, dual-head U-Net, PatchGAN discriminator, speckle preservation loss.
Tech: PyTorch, Lightning, Sentinel-1/2, VGG19, EMA, TTUR
- **Talent Insider – Job Platform:** Developed a scalable recruitment system supporting **1K+ users** with secure bcrypt-based authentication and live job CRUD operations. Engineered RESTful APIs and optimized DB queries to be sub-100ms.
Highlights: Modular MVC structure, asynchronous API calls, and transactional MySQL operations for data consistency.
Tech: Java, Spring Boot, JDBC, MySQL, Maven, Java Swing

LEADERSHIP & CAMPUS ROLES

- **Google Developer Groups (GDG) Bennett University** 2025 – 2026
Co-Organizer
 - Led and organized developer-focused workshops, hackathons, and technical talks on AI, ML, cloud, and system design, collaborating with industry speakers and scaling student developer engagement.
- **SPARK, Entrepreneurship Cell** 2025
Joint Secretary
 - Drove entrepreneurship initiatives, coordinated cross-functional teams for flagship events, and supported sponsorship outreach and startup engagement activities.
- **Rotary International – Youth Advisory Council (USA)** Jul 2023 – Jun 2024
Board Member
 - Collaborated with global youth leaders to shape Rotary’s youth exchange programs and contributed to international policy discussions.